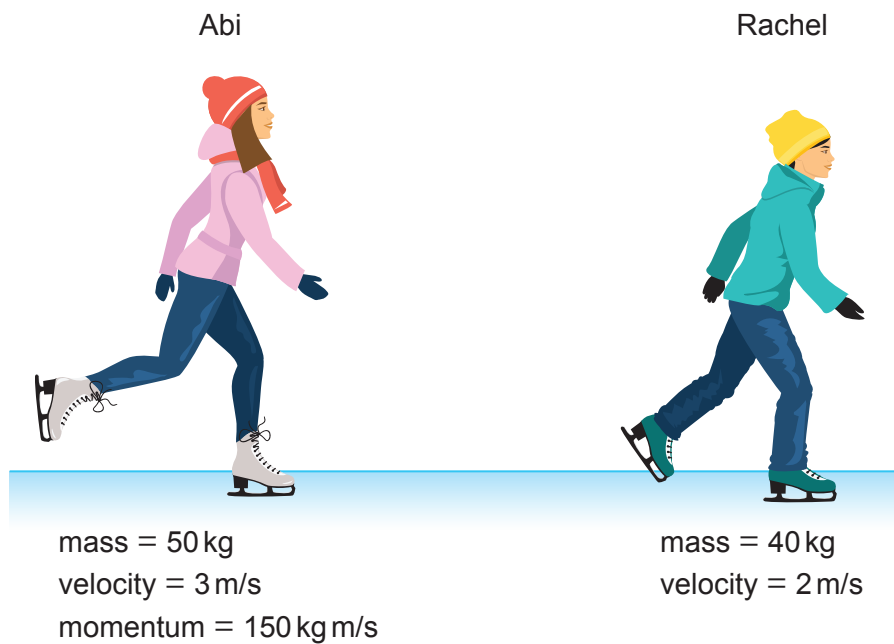


5. The diagram below shows two ice skaters, Abi and Rachel, skating in the same direction.



- (a) (i) Use the diagram above and the equation:

$$\text{momentum} = \text{mass} \times \text{velocity}$$

to calculate **Rachel's** momentum.

[2]

Rachel's momentum = ..... kg m/s

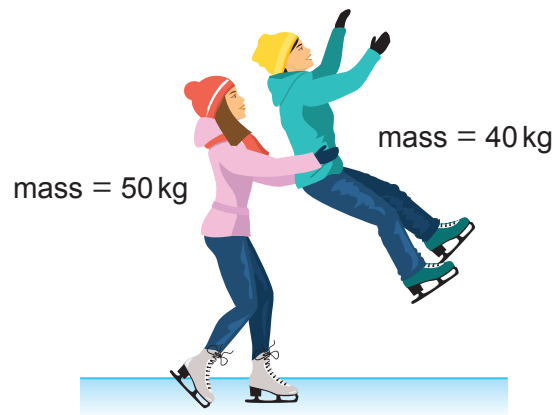
- (ii) Use your answer to (a)(i) and information from the diagram to determine the total momentum of the ice skaters.

[1]

total momentum = ..... kg m/s



- (iii) Abi catches up with Rachel and lifts her as shown below. This can be thought of as a collision.



State the total momentum of the ice skaters after this collision has taken place. [1]

total momentum = ..... kg m/s

- (iv) Use the equation:

$$\text{velocity} = \frac{\text{total momentum}}{\text{mass}}$$

to calculate the velocity of the ice skaters after the collision. [2]

velocity = ..... m/s

- (v) The ice skaters glide freely from this velocity,  $u$ . After a time,  $t$ , of 25 s their velocity,  $v$ , is 0 m/s.

Use the equation:

$$x = \frac{1}{2} (u + v)t$$

to calculate the distance,  $x$ , that they travel. [2]

$x = \dots\dots\dots$  m



- (b) Abi suggests that if she doubles her velocity the total momentum of Abi and Rachel will also double.  
Explain whether Abi is correct. [2]

.....

.....

.....

Examiner  
only

10

